SymPy Bug Card

Bug 5: antiderivative has the wrong sign for negative x

Task. Compute an indefinite integral:

$$\int \frac{\sqrt{x^2+1}}{x}\,dx.$$

SymPy's answer. SymPy returns

$$\frac{x}{\sqrt{1+x^{-2}}} - \operatorname{asinh}\left(\frac{1}{x}\right) + \frac{1}{x\sqrt{1+x^{-2}}}.$$

Correct answer. The returned formula differentiates to

$$\sqrt{1+x^{-2}},$$

which differs from the original integrand $\sqrt{x^2+1}/x$ on the negative real interval.

Explanation. An antiderivative must differentiate back to the integrand. Here, substituting $x=-2$ after differentiation gives $+\sqrt{5}/2$, while the original integrand gives $-\sqrt{5}/2$.

Likely root cause. The diagnosis locates the branch loss in `sympy/integrals/meijerint.py`, where the Meijer-G indefinite integration path rewrites through an x^2 argument and then denests powers, producing a positive-branch formula without the needed $\operatorname{sign}(x)$ factor. This is a new instance of a known failure mode.

Artifact (GitHub).
`bug-report/bug-005-antiderivative-has-the-wrong-sign-for-negative-x/`